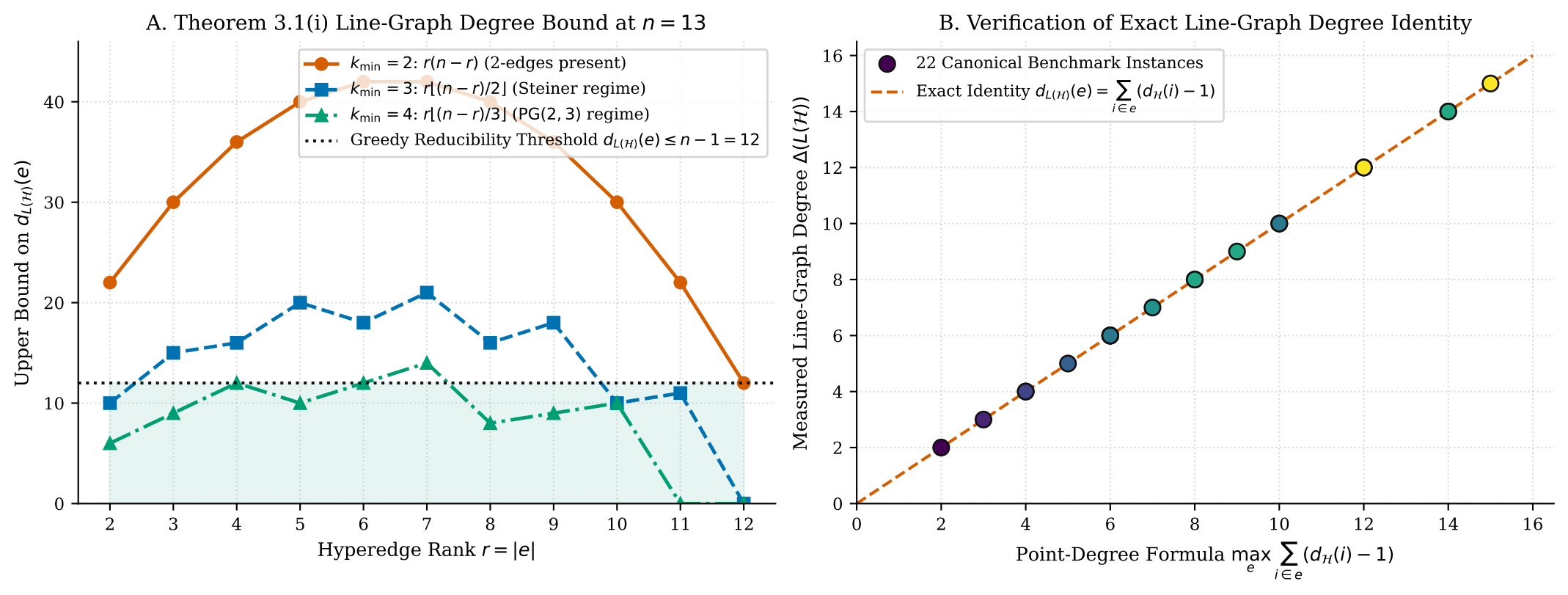
A. Theorem 3.1(i) Line-Graph Degree Bound at $n = 13$
$k_{\min} = 2$: $r(n-r)$ (2-edges present)
$k_{\min} = 3$: $r\lfloor (n-r)/2 \rfloor$ (Steiner regime)
$k_{\min} = 4$: $r\lfloor (n-r)/3 \rfloor$ (PG(2, 3) regime)
Greedy Reducibility Threshold $d_{L(\mathcal{H})}(e) \le n - 1 = 12$
Upper Bound on $d_{L(\mathcal{H})}(e)$
Hyperedge Rank $r = |e|$
B. Verification of Exact Line-Graph Degree Identity
22 Canonical Benchmark Instances
Exact Identity $d_{L(\mathcal{H})}(e) = \sum_{i \in e} (d_{\mathcal{H}}(i) - 1)$
Measured Line-Graph Degree $\Delta(L(\mathcal{H}))$
Point-Degree Formula $\max_e \sum_{i \in e} (d_{\mathcal{H}}(i) - 1)$